ECON 370: Assignment 3

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Due: Before class on October 28th, 2025

Directions

As a reminder, LLMs and AIs are allowed from use on this assignment.

This assignment serves dual purpose: **application of `tidyverse`** and **cleaning and merging some of your data for the final project**. You can check **final project details** as a guide. You can use AI but consider using StackOverflow before asking AI for a solution!

Use the Rmd template and write code to answer the following questions. In order to ensure smooth grading, I will ask you to write both your code and explanation (as comment) in code chunk. Please write your code and answer in the corresponding place.

Each question is worth three points. For each section, a "question" corresponds to a number.

Please **comment** your code clearly. 0.5 point per question will be deducted if there is a significantly lack of commenting. Also, the easier your code is to read, the more likely I will be able to figure out what you're doing.

Remember to **host your data on GitHub** if it cannot directly be pulled using functions such as `tq_get`.

There are many ways to clean data. However, I will ask you to apply **pipeline and `dplyr` package** for data wrangling here. 0.5 point per question will be deducted if other method is used while getting the correct answer.

You need to work individually for this homework.

O. Return Data

This is the code to extract return data online. Do not modify.

Note: we are using the S&P 500 index returns, not individual companies in the S&P 500.

I. Variables and Reasoning

In this assignment, I will ask you to pull **three variables each from a different source** (you will need to do at least 10 from at least 4 sources in the final project).

1. Name the three variables, sources and frequency of your data.

2. Reason to include each variable(articles or strong logic support).

3. Upload all your **raw** data on GitHub unless they can be extracted using R functions (without downloading things). Make sure they have **long enough history**. Then download each data as a **tibble**:

II. Data Cleaing

1. Check the **date format** of your datasets. Do you see different formats than "yyyy-mm-dd"? Do you different ways of documenting with month end? If so, fix them with **`mutate`** function combined with `lubridate` package.

2. Use `left_join` to **join** all your data with the return data "SP500_simple_returns".

3. What is the situation of **mixed frequency** and **missing data**? If there are, fill in with appropriate method. One reference of mixed frequency is MIDAS.

4. What is the situation of **publication bias** for all three variables? Are there any? If there are, use `mutate` and `lag` to make appropriate lags.

5. Use `lead` to create predictive date and predictive return.

III. Exploration with `tidyverse`

1. Make the data longer using `pivot_longer`, each variable name are stored in a column called "Var_name" and all values are stored in a column called "Var_value".

2. Use `group_by` to group by different values in "Var_name" and estimate the **mean** for each variable using `summarise`.

3. Use **ggplot** to visualize all the series together in a single plot. Examine (with the plot) whether the series fluctuate around a stable level or display a trend that drifts away over time.

Note: if you find one of your varaible displays a trend that drifts away over time, you should reconsider how to use that variable. (This is called non-stationary series in time series.)